

Arjuna JEE 2025

Maths

DPP: 5

Basic Mathematics

Q1 Solution of inequality $2x - 3 > 0$ and $4 - 3x < 0$.

- (A) $x > \frac{4}{3}$
 (B) $x < \frac{4}{3}$
 (C) $x > \frac{3}{2}$
 (D) $x < \frac{3}{2}$

Q2 Values of 'x' satisfying both inequalities:

$$x - 3 \leq 4 \text{ \& } \frac{x-5}{x+1} > 0$$

- (A) $x \in (-\infty, 7]$
 (B) $x \in (-\infty, -1)$
 (C) $x \in (-\infty, -1) \cup (5, 7]$
 (D) $x \in (5, 7]$

Q3 Solution of $\frac{2x-x^2}{(3+x)(5-x)} < 0$ contains smallest negative integral value is:

- (A) -3 (B) -2
 (C) -1 (D) 0

Q4 Values of 'x' satisfying: $-1 < \frac{2x}{x-1} \leq 4$

- (A) $x \in (-\infty, \frac{1}{3}) \cup (1, \infty)$
 (B) $x \in (-\infty, \frac{1}{3}) \cup (2, \infty)$
 (C) $x \in (-\infty, \frac{1}{3}) \cup [2, \infty)$
 (D) $x \in (-\infty, \frac{1}{3}) \cup [2, \infty) \cup \{1\}$

Q5 Solution of inequality: $2x^2 - 2x - 1 < 0$

- (A) $x \in \left(\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$
 (B) $x \in \left[\frac{1-\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right]$
 (C) $x \in \left(-\infty, \frac{1-\sqrt{3}}{2}\right) \cup \left(\frac{1+\sqrt{3}}{2}, \infty\right)$
 (D) $x \in (-1, 2)$

Q6

All real set of values of x satisfying $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \leq 0$ are given by $[a, b] \cup (c, d]$ then value of $\frac{a+d}{c-b}$ is equal to _____.

Q7 Solve $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$

Q8 Solve $\frac{5-4x}{3x^2-x-4} < 4$

Q9 Solve $\frac{20}{(x-3)(x-4)} + \frac{10}{x-4} + 1 > 0$

Q10 If $x = \frac{4}{(\sqrt{5}+1)(\sqrt[4]{5}+1)(\sqrt[8]{5}+1)(\sqrt[16]{5}+1)}$. Then the value of $(1+x)^{48}$ is-

- (A) 5 (B) 25
 (C) 125 (D) 625

Q11 If $a + b + c = 0$ then $x^{\frac{a^2}{bc}} \cdot x^{\frac{b^2}{ac}} \cdot x^{\frac{c^2}{ab}}$ is equal to

- (A) x (B) x^2
 (C) 1 (D) x^3

Q12 If $\sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c} = 0$ and $(a+b+c)^3 =$

- (A) abc
 (B) $3abc$
 (C) $9ac$
 (D) $27abc$

Q13 The complete solution of $\frac{x^2-1}{x+3} \geq 0$ & $x^2 - 5x + 2 \leq 0$ is:

- (A) $x \in \left[\frac{5-\sqrt{17}}{2}, \frac{5+\sqrt{17}}{2}\right]$
 (B) $x \in \left[1, \frac{5+\sqrt{17}}{2}\right]$
 (C) $x \in (-3, -1]$



(D) $x \in (-3, -1] \cup [1, \infty)$



Answer Key

Q1 (C)

Q2 (C)

Q3 (B)

Q4 (C)

Q5 (A)

Q6 2

Q7 $x \in [1, 3] \cup [5, \infty)$

Q8 $x \in \left(-\infty, -\frac{\sqrt{7}}{2}\right) \cup \left(-1, \frac{\sqrt{7}}{2}\right) \cup \left(\frac{4}{3}, \infty\right)$

Q9 $x \in (-\infty, -2) \cup (-1, 3) \cup (4, \infty)$

Q10 (C)

Q11 (D)

Q12 (D)

Q13 (B)



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